

COMP CO 835

Object-Oriented Systems

Week 9

Designing the Database

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Outline

- Databases and Database Management Systems
- Database Design and Administration
- Relational Databases
- Distributed Database Architectures
- Protecting the Database
- Integrating Databases into systems development
- Data Administrators & Database Administrators in SDLC.
- Data Security in I.T Project Management

Learning Objectives

- Explain the responsibilities of the data administrator and database administrator
- Design a relational database schema based on a class diagram
- Evaluate and improve the quality of a database schema
- Describe the different methods for configuring distributed databases
- Explain the importance of and methods for protecting the database

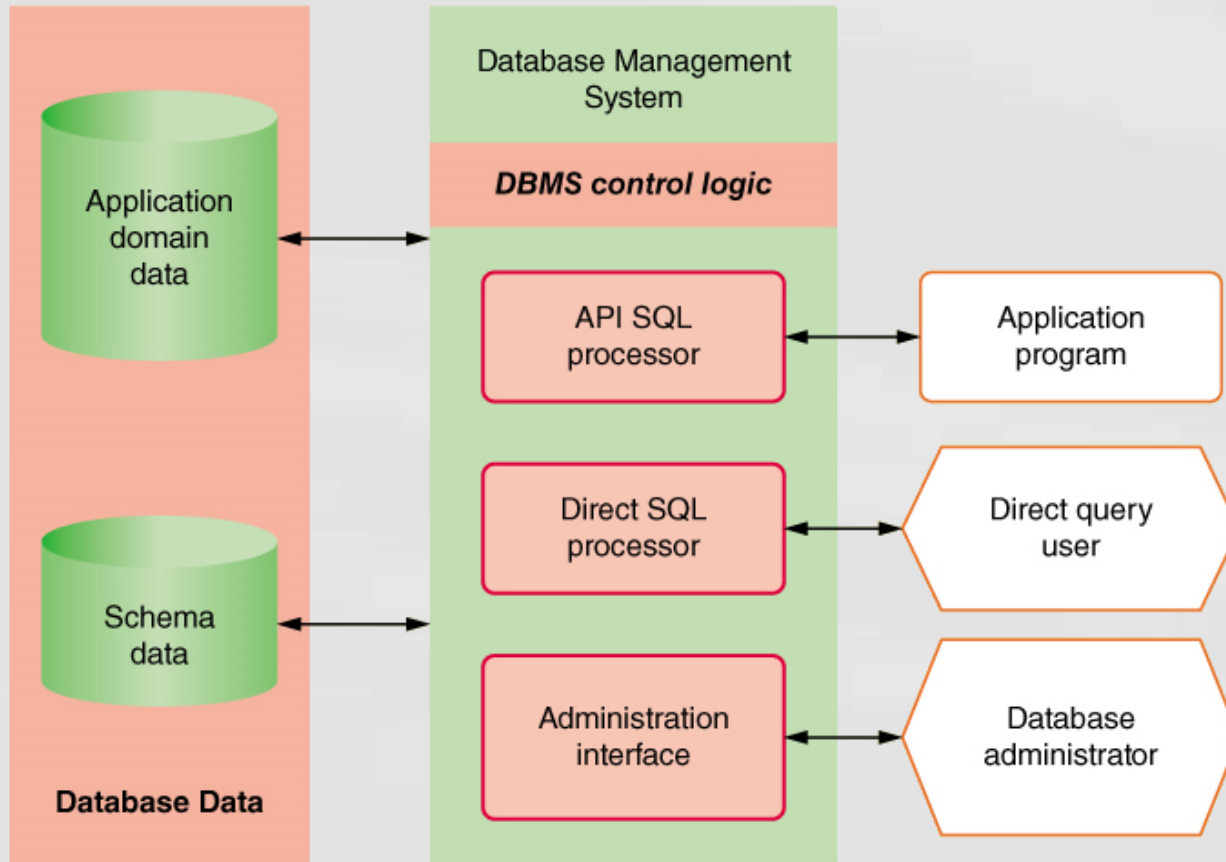
Databases and Database Management Systems

- Databases and database management systems are important components of a modern information system
- Database design transforms the domain model class diagram into a detailed database model for the system
- A database is an organized collection of data or information stored electronically in a computer system
- A database management system is used to implement and interact with the database

Databases and DBMSs

- Database (DB) -- an integrated collection of stored data that is centrally managed and controlled
- Database management system (DBMS) -- a system software component that manages and controls one or more databases
 - MySQL, Microsoft Access, SQL Server, FileMaker, Oracle, dBASE, etc are examples of DBMSs.
 - DBMSs have a way they communicate with each other
- Schema -- database component that contains descriptive information about the data stored in the physical data store (sometimes called *metadata*)
- Structured Query Language -- the standard query language to access and update data in a relational DBMS

DBMS Components



Database Schema

- Organization of individual stored data items into higher level groups, such as tables
- Associations among tables or classes
- Details of physical data store organization, including types, lengths, locations, and indexing of data items
- Access and content controls, including allowable values for specific data items, value dependencies among multiple data items, and lists of users allowed to read or update data items

Databases

- How does database design integrate within the project plan?
 - A complete information system comprises people, hardware, software, the database(s)/DBMS, application programs & procedures.
- The database is part of a larger whole information system, which provides for data collection, storage and retrieval.
 - Water-fall development – design and implement database first
 - Iterative development – database is foundational, early iterations need to focus on data and key portions of the database.
- Iterative development – important to consider database impacts of all subsystems

Characteristics of a DBMS

- Simultaneous access by many users and many applications
- Direct access to data with a data interface
- Uniform and consistent access
- Integration and distribution of data across multiple servers
- Most DBMSs comes with an Open Database Connectivity driver that allows the database to integrate/communicate with other databases.

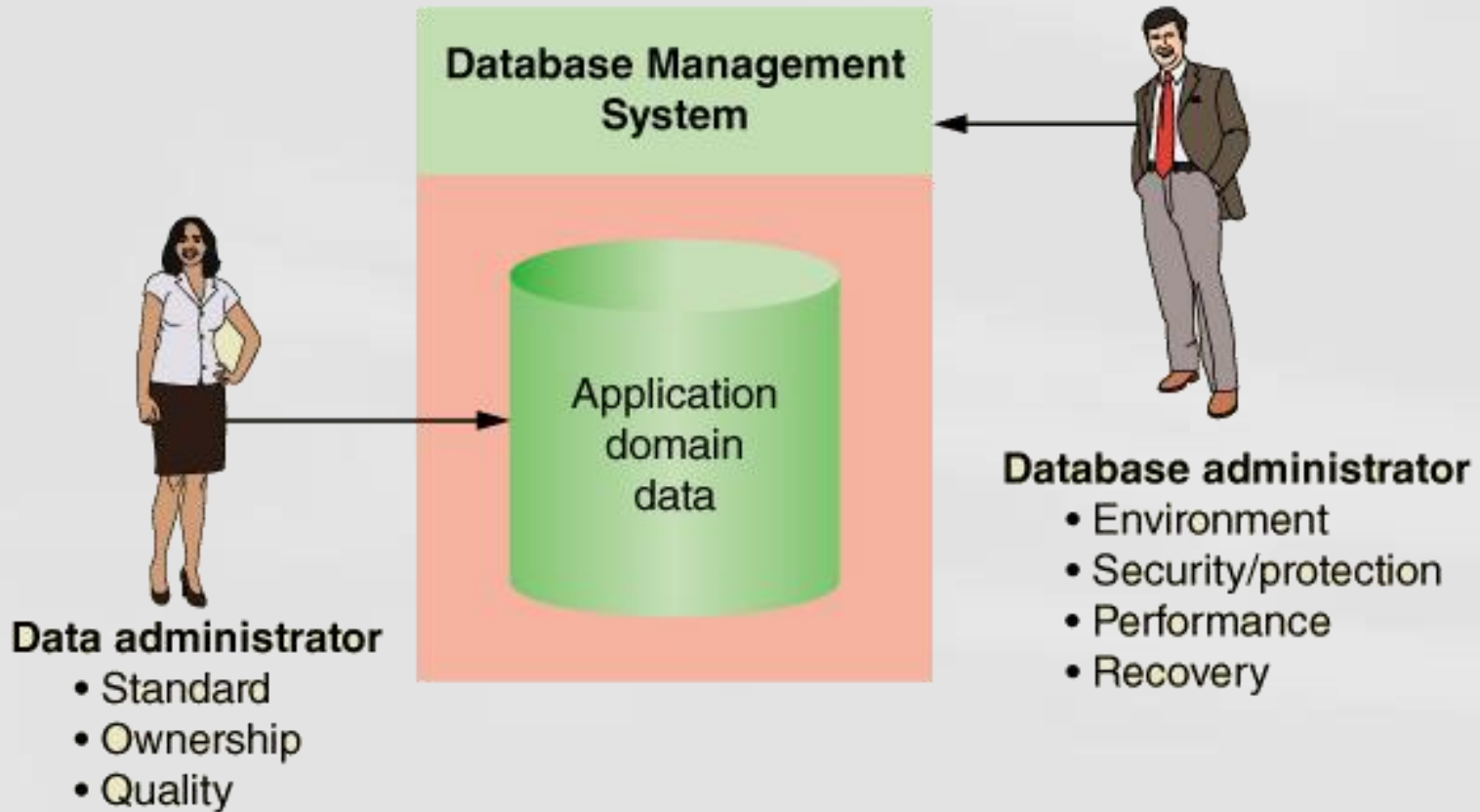
Database Design and Administration

- Who is involved in database design?
- Data Administrator (DA) – person in charge of structure and integrity of the data
 - Data standards – naming, definition, data typing
 - Data use – ownership, accessibility, confidentiality
 - Data quality – validation rules, completeness, currency

Database Design and Administration

- Who is involved in database design?
- Database Administrator (DBA) – person in charge of safety and the operation of the database
 - Manage multiple DBMS environment
 - Protect the database and data – authentication
 - Maintain high-performance level
 - Backup data and define recovery procedures

DA and DBA Responsibilities



Software Development & Databases

- **Databases in Project Management/SDLC**
 - Important because they hold the data and information for a business or project
- **Computer Application, and Information Systems (Require Databases)**
- **Information system** – a set of interrelated components that collects, processes, stores, and provides as output the information needed to complete business tasks
 - Broader in scope than “app”
 - Includes database and related manual processes
- **An Information System cannot function without a database**
 - Without a database or database management system, running and managing data effectively in a business or project is not possible.
- Databases are very important to every business because they store and communicate information related to business transactions, product inventory, customer profiles, marketing activities, and overall business functions.

Relational Databases (1 of 2)

- Relational database management system (RDBMS) -- a DBMS that organizes data in tables (relations)
- Table -- a two-dimensional data structure of columns and rows
- Row -- one horizontal group of data attribute values
- Attribute -- one vertical group of data attribute values
- Attribute value -- the value held in a single table cell

Relational Databases (2 of 2)

- Key – an attribute or set of attributes, the values of which occur only once in all the rows of the table
- Candidate Key – an attribute or set of attributes that could serve as the primary key
- Primary key – the key chosen by a database designer to represent relationships among rows in different tables
 - It is a unique identifier for each record in a table, a specific field (or set of fields) chosen to ensure that each row in the table is uniquely identifiable.
 - Example, in a Users table, a user_id field could serve as the primary key, uniquely identifying each user, while other tables (like Orders) might reference this user_id as a foreign key to link specific orders to
- Foreign key – an attribute that duplicates the primary key of a different (or foreign) table

Partial Display of a Relational Database Table

Field or attribute names

One row, tuple, or record

One field or attribute value

One field or attribute and its values

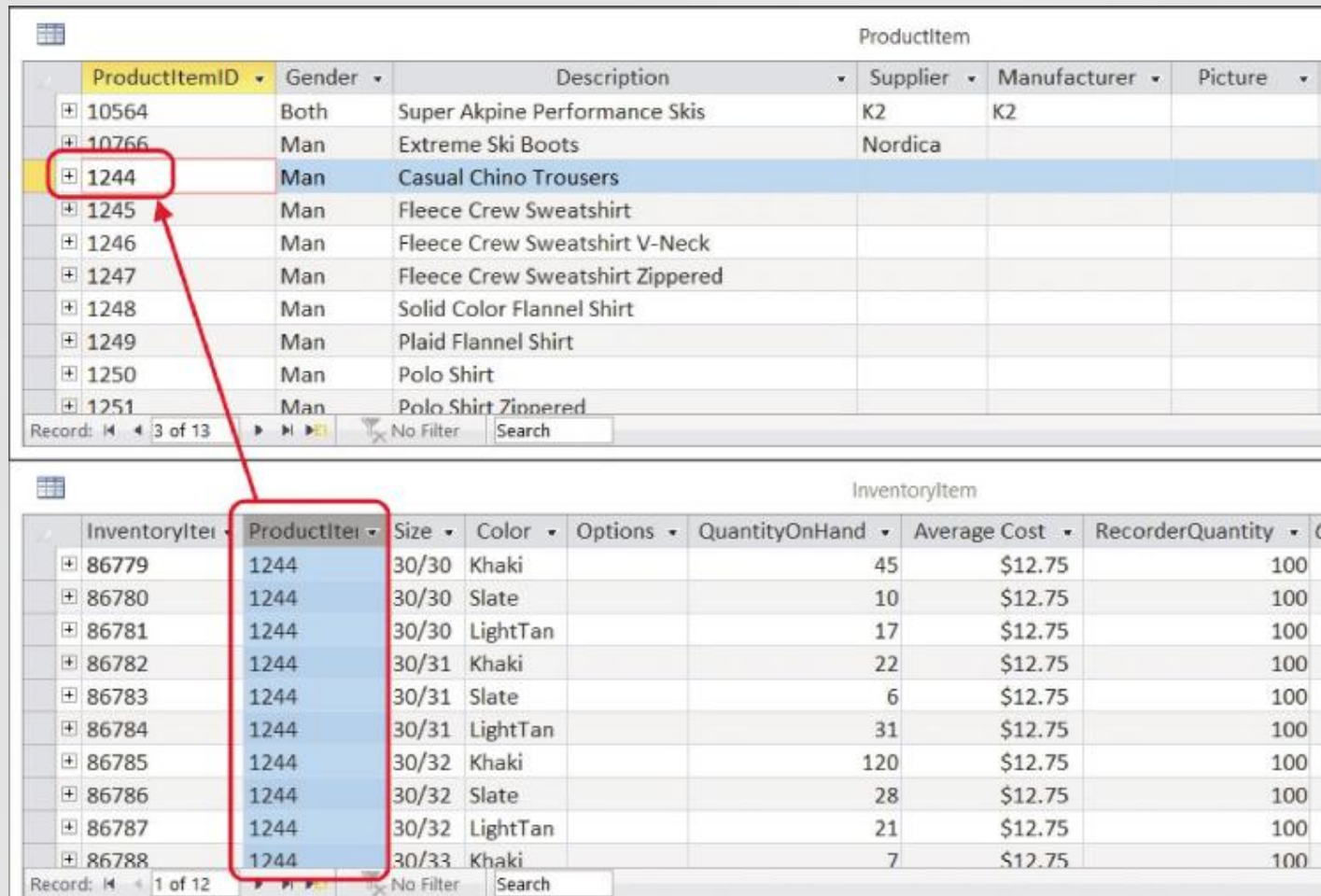
ProductItemID	Gender	Description	Supplier	Manufacturer	Picture
10564	Both	Super Akpine Performance Skis	K2	K2	
10766	Man	Extreme Ski Boots	Nordica	Nordica	
1244	Man	Casual Chino Trousers	West Coast	Adida	
1245	Man	Fleece Crew Sweatshirt	West Coast	Adida	
1246	Man	Fleece Crew Sweatshirt V-Neck	West Coast	Adida	
1247	Man	Fleece Crew Sweatshirt Zippered	West Coast	Adida	
1248	Man	Solid Color Flannel Shirt	RMO	RMO	
1249	Man	Plaid Flannel Shirt	RMO	RMO	
1250	Man	Polo Shirt	RMO	RMO	
1251	Man	Polo Shirt Zippered	RMO	RMO	
1252	Man	Navigator Jacket	Colorado Supply	North Face	
1253	Man	Navigator Jacket Hooded	Colorado Supply	North Face	
1254	Man	Cotton Thermal Shirt	Colorado Supply	Under Armour	

Record: 3 of 13

No Filter

Search

An Association Between Rows in Two Tables (key and foreign key)



The image displays two database tables from a software application. The top table, titled "ProductItem", lists various products with columns: ProductItemID, Gender, Description, Supplier, Manufacturer, and Picture. The bottom table, titled "InventoryItem", lists inventory items with columns: InventoryItemID, ProductItemID, Size, Color, Options, QuantityOnHand, Average Cost, and RecorderQuantity. A red box highlights the ProductItemID 1244 in the top table, and a red box highlights the corresponding ProductItemID 1244 in the bottom table. A red arrow points from the top box to the bottom box, illustrating the foreign key relationship.

ProductItemID	Gender	Description	Supplier	Manufacturer	Picture
10564	Both	Super Alpine Performance Skis	K2	K2	
10766	Man	Extreme Ski Boots	Nordica		
1244	Man	Casual Chino Trousers			
1245	Man	Fleece Crew Sweatshirt			
1246	Man	Fleece Crew Sweatshirt V-Neck			
1247	Man	Fleece Crew Sweatshirt Zippered			
1248	Man	Solid Color Flannel Shirt			
1249	Man	Plaid Flannel Shirt			
1250	Man	Polo Shirt			
1251	Man	Polo Shirt Zippered			

InventoryItemID	ProductItemID	Size	Color	Options	QuantityOnHand	Average Cost	RecorderQuantity
86779	1244	30/30	Khaki		45	\$12.75	100
86780	1244	30/30	Slate		10	\$12.75	100
86781	1244	30/30	LightTan		17	\$12.75	100
86782	1244	30/31	Khaki		22	\$12.75	100
86783	1244	30/31	Slate		6	\$12.75	100
86784	1244	30/31	LightTan		31	\$12.75	100
86785	1244	30/32	Khaki		120	\$12.75	100
86786	1244	30/32	Slate		28	\$12.75	100
86787	1244	30/32	LightTan		21	\$12.75	100
86788	1244	30/33	Khaki		7	\$12.75	100

Designing Relational Databases: Based on the Domain Model Class Diagram

1. Create a table for each class
2. Choose a primary key for each table (invent one, if necessary)
3. Add foreign keys to represent one-to-many associations
4. Create new tables to represent many-to-many associations
5. Represent classification hierarchies
6. Define referential integrity constraints
7. Evaluate schema quality and make necessary improvements
8. Choose appropriate data types
9. Incorporate integrity and security controls

Rows					
Columns					

- **Rows** – Horizontal – From Left to Right
- **Columns** – Vertical – From Top to bottom (up to down)

Representing Associations

- One-to-Many – Add primary key attribute of the “one” class to the “many” class as a foreign key
- Many-to-Many –
 - With an Association Class – Add primary keys of endpoint classes as foreign keys and as candidate keys. May also become primary key
 - Without an Association Class – Create new table. Add primary keys of endpoint classes as foreign keys and as candidate keys.

Designing Relational Databases

- Referential integrity -- a consistent state among foreign key and primary key values
- Referential integrity constraint -- a constraint, stored in the schema, that the DBMS uses to automatically enforce referential integrity

Functional Dependency (1 of 2)

- A relationship between attributes such that the values in the first attribute (or set) always determine the values in the second attribute (or set)
- *Attribute B is functionally **dependent** on attribute A if for each value of attribute A there is only one corresponding value of attribute B.*
 - *Written as FD: $A \rightarrow B$.*
 - *Also stated as A functionally **determines** B*

Functional Dependency (2 of 2)

- ProductID → Supplier
- But **NOT** Supplier → ProductID



The screenshot shows a database table named 'ProductItem' with the following columns: ProductItemID, Gender, Description, Supplier, and Manufacture. The table contains 13 records. The record with ProductItemID 1244 is highlighted in blue. The table demonstrates a functional dependency where ProductItemID determines the Supplier, but the Supplier does not determine the ProductItemID.

ProductItemID	Gender	Description	Supplier	Manufacture
10564	Both	Super Akpine Performance Skis	K2	K2
10766	Man	Extreme Ski Boots	Nordica	Nordica
1244	Man	Casual Chino Trousers	West Coast	Adida
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First, Second, and Third Normal Form

- Normalization is the process of producing high-quality databases without update, insertion, or delete anomalies
- Database Normalization

Data Types

- The data type defines the storage format and allowable content of an attribute (field)
- Primitive data types – data types supported directly by the DBMS
- Complex data types – combinations or compositions of primitive data types. User defined

Standard Primitive Data Types

Type(s)	Description
datetimeoffset	Date, time, and time zone
int, small int, and bigint	Whole numeric values
float and real	Numeric values with fractional quantities
money	Currency values and related symbols (e.g., \$ and €)
nchar and nvarchar	Fixed- and variable-length Unicode string
varbinary	Variable-length byte sequence up to 2GB
xml	XML document up to 2GB

Distributed Database Management System

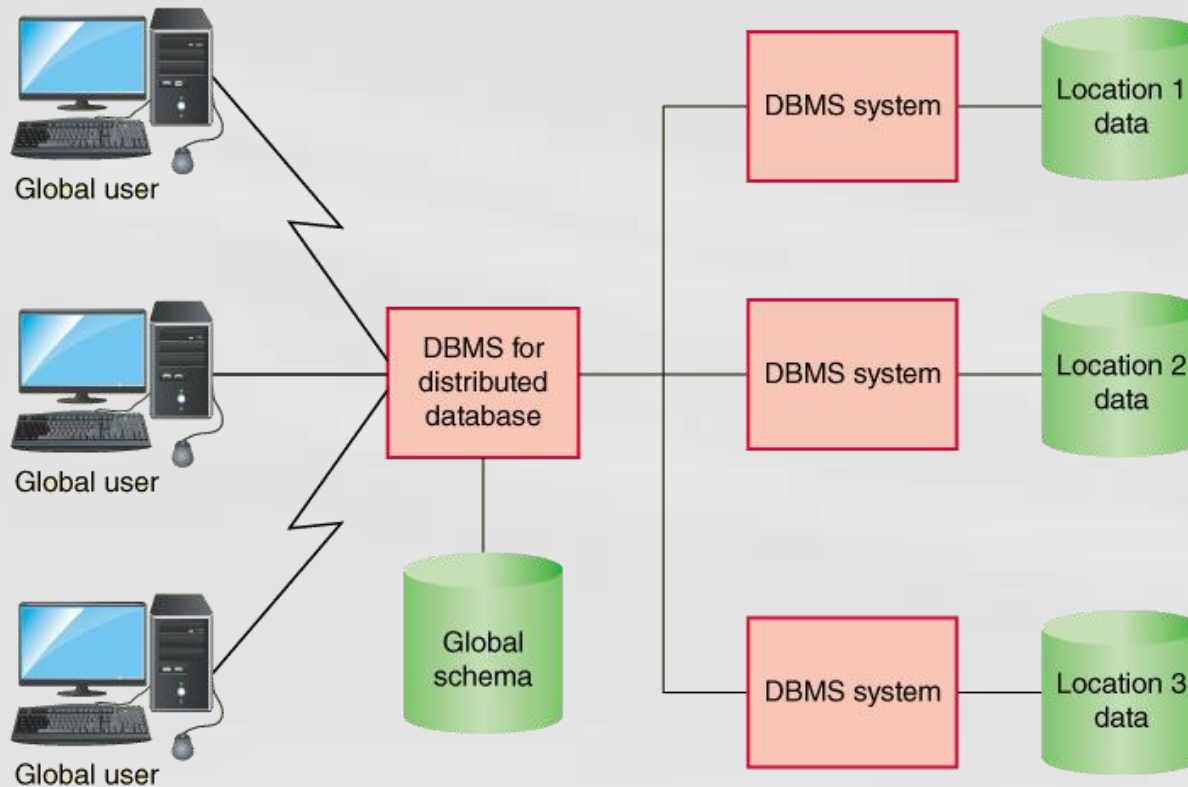
- A **distributed database** is a database in which data is stored across different physical locations.
- It is a collection of multiple interconnected databases across various locations that communicate via a computer network.
- Features of a Distributed Database
 - Databases in the collection are logically interrelated with each other.
 - Data is physically stored across multiple sites. Data in each site can be managed by a DBMS independent of the other sites.
 - The processors in the sites are connected via a network.
 - A distributed database incorporates transaction processing
- Distributed Database Management System

Distributed Database Architectures

- Decentralized database is stored at many locations but not requiring interconnectivity or synchronization
- Homogeneous distributed database is stored at multiple locations, with all locations using the same DBMS. Coordinated with a global schema
- Heterogeneous distributed database is stored at multiple locations and with different DBMS and may have local schemas.

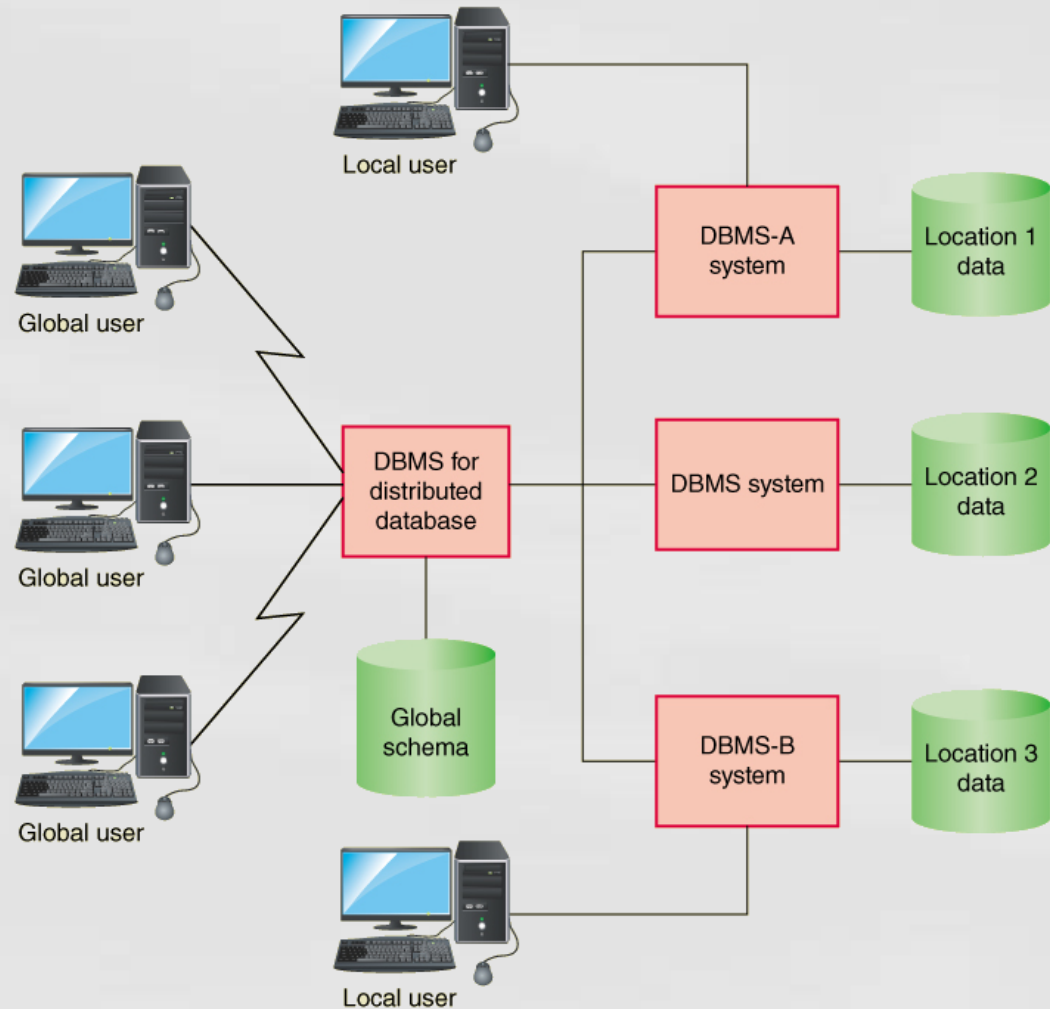
Homogeneous Distributed Database

- Access is through a common DBMS and schema



Heterogeneous Distributed Database

- Access is through distinct DBMSs. May have global and local schemas in operation



Protecting the Database (1 of 2)

- Transaction Logging – a technique to record all updates including change, date, time, user
 - Helps to prevent fraud
 - Recovery mechanism for failures
- Every database has a transaction log that records all transactions and the database modifications made by each transaction.
- The transaction log is a critical component of the database. If there is a system failure, you will need that log to bring your database back to a consistent state.

Protecting the Database (2 of 2)

- Concurrency and Update Controls

- Transaction – a piece of work with several steps, either all must complete or none must be accepted
- Database lock – technique to apply exclusive control to a portion of the database to one user at a time
- Shared or read lock – a lock where multiple transactions (users) may read the data
- Exclusive or write lock – a lock where only one transaction (user) may access the locked portion of the database

Protecting Data & Information

- Data Security

- Is the process of protecting digital data, such as those in a database, from destructive forces and from unauthorized users, such as hackers, a data breach or cyberattack.
- Data security is the prevention of unauthorized access, use, disruption, modification or destruction of data in storage.

- Information Security

- Information security (InfoSec), is the practice of protecting information by mitigating information risks.
- It is the prevention of unauthorized access, use, disruption, modification or destruction of information

- Data security is a layer of Information Security. Data is a part, and information is the whole.

Data is a collection of facts, while information puts those facts into context and makes meaning out of the data. **Information** is the result of analyzing and interpreting pieces of data.

Protecting Data & Information

- **The CIA (Confidentiality, Integrity, & Availability)**
 - Database security should maintain confidentiality, integrity, and availability of data and information.
 - Required for overall security in a database system.



Summary

- Most modern information systems store data and access data using a database management systems (DBMS)
- The most common database model is a relational database (RDBMS), which is a collection of data stored in tables
- The relational database schema is developed based on the domain model class diagram Each class is represented as a table. One to many associations are represented by adding foreign keys
- Normalization is the process to produce high-quality databases without update, insertion or delete anomalies
- Distributed databases are necessary for very large databases
- Data Administrator (DA) – person in charge of structure and integrity of the data.
- Database Administrator (DBA) – person in charge of safety and the operation of the database.

Summary

- Databases are important in systems development projects
- Data is a collection of facts, while Information is the result obtained from analyzing and interpreting the data.
- Data security is the prevention of unauthorized access, use, disruption, modification or destruction of data in storage.
- Information security is the prevention of unauthorized access, use, disruption, modification, or destruction of information.
- The Pillars of Security are Confidentiality, Integrity, Availability, Authenticity, Authorization, and Non-repudiation.
- The Three major pillars of Information Security are Confidentiality, Integrity, & Availability, known as the CIA.
- The protection of data and information is one of the major things that must be considered by the project team when designing and developing Information Systems.

Questions?